Power Rangers - Net Energy Generation by State by Month

Team Members:

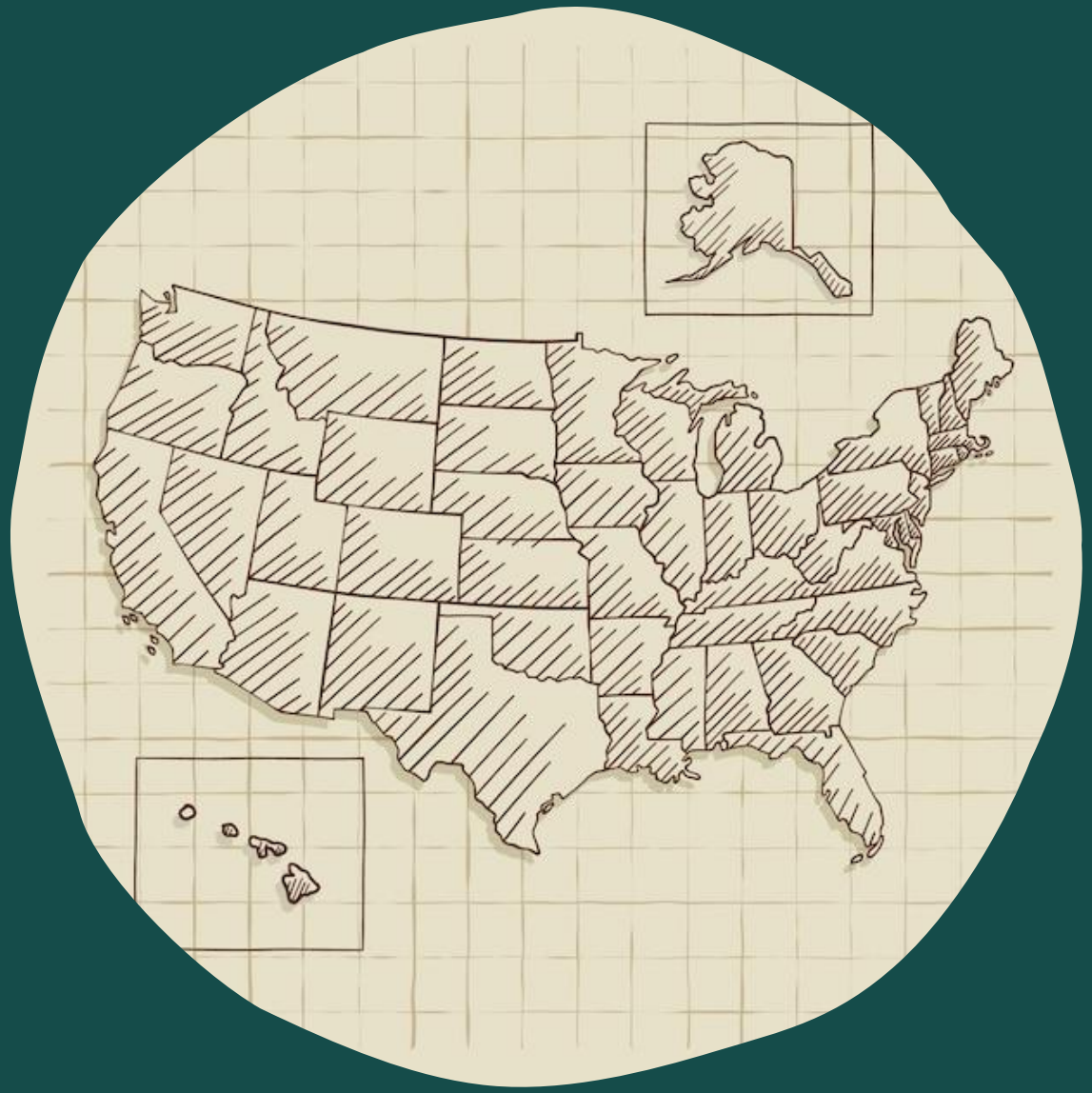
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Project Category: Energy

Problem Statement:

Energy generation and capacities are not clearly identifiable. This makes it difficult to understand where electricity comes from and how much can be produced. Utility companies need this information to compare energy sources and make informed decisions about energy planning and future energy investments.

A dashboard and interactive U.S. map will help identify generation capacity and the power sources contributing to consumers' energy supply. A SQL database will organize states, energy sources, and monthly generation data, allowing utility companies to compare states and time periods. As the number of users increases, efficient database queries and additional hosting resources can help maintain performance.



Project Plan

GitHub Repo: https://github.com/mkuhn23/Power-Rangers_4723Project

Data Source: https://www.eia.gov/electricity/monthly/

- Dataset: Appendix C (Electric Power Data by Month and State, 2001 to the Present,xls)

Approach:

- Clean selected portion of workbook and import as SQL database for states, energy sources, and monthly generation.
- We will calculate totals and compare the generation mixes across states at different time periods.
- Using the Streamlit Python package to create a dashboard and official U.S. map shapefiles, we will package the app and dependencies with Docker so all members can run the environment.
- GitHub will be used to manage the project, and all members will be working on individual branches to maintain version control and checks before individual work is merged.

Expected Capabilities: Comparison between states, comparison between energy sources, high-level access for non-SMEs through map, understanding of U.S. energy “Power States”.

Timeline and Potential Improvements

- By Oct. 15:
  - Create full SQL database
  - Full power generation calculations
  - Entire GitHub structure and Docker environment created
  - Choose graphics for comparison and visualization options
- By Oct. 30:
  - Create Streamlit dashboard structure and base layer of Python code
  - Working backup of the interactive U.S. Map
  - Start integrating data into the dashboard and test functionality
- By Nov. 15:
  - Full working dashboard with map integrations
  - Slide show that shows path to success
  - Report Started

Scope for Improvement: Add energy generation forecast dataset, consumption layer that shows differences between generation vs. consumption and potential bottlenecks